

RMS & Admin Platform

Progress Report

One working day turning the RMS Admin dashboard from a single working button into a spec-complete risk, entity, and performance layer across every role.

DATE	BRANCH	COMMITTS TODAY	SPEC REFERENCE
24 August 2026	feature/dev	3, on feature/dev	iwmQT GUI Spec v1.0

What shipped today

Four phases, built and verified in sequence: every role dashboard wired end to end, a real multi-tenant entity model, all 14 RMS pre-trade risk controls, and the HFT-grade grid/WebSocket layer the spec calls for.

38

files changed since yesterday

7

new database tables / extensions

14 / 14

RMS pre-trade controls covered

5 / 5

role dashboards fully wired

- **RMS Admin dashboard completed** — Kill Switch, OMS/Risk Config, Audit Log, and User & Role Management went from dead cards to working, audited features.
- **Company Account is now a real entity** — a proper companies table backs it, with Super-Admin-driven create/suspend and entity-scoped user management, not just a role label.
- **All 14 RMS pre-trade controls are enforced** on every order placement, each with its own labelled rejection message.
- **Grid and market-data layer rebuilt for HFT** — row-level transaction updates, a real WebSocket delta protocol with sequence numbers, CSV export, column persistence, and per-role code-splitting.

How the day went

Four sequential phases, each scoped and verified before moving to the next.

07:48

● **Baseline**

Starting point — RMS & Super Admin dashboard shells existed; only the "RMS Dashboard" card was functional anywhere.

Morning

● **Phase 1 — RMS Admin dashboard completed**

Kill Switch, OMS Risk Config, Audit Log, and User & Role Management built and wired in. Checkpointed at 10:38 as 4eec107.

```
commit 4eec107
```

Midday

● **Phase 2 & 3 — Every role wired, entity model, risk controls**

Gap-analysed the full GUI spec, then wired PM / Company Account / Super Admin dashboards, built the Company entity model, and implemented all 14 RMS pre-trade controls plus LIMIT-only order entry and the OPS rate limit. Committed 14:25 as d2d95cc.

```
commit d2d95cc
```

Afternoon

● **Phase 4 — HFT grid & WebSocket rework**

Rebuilt the market-data WebSocket as a true delta protocol, converted every grid to row-level transactions, added CSV export, column persistence, and split the app bundle by role. Verified and committed.

commit 634ca3c

RMS Admin dashboard & role wiring

Every module card, across all five roles, now does something real — reusing one shared Admin Portal instead of five one-off screens.

Kill Switch DONE

Global and per-user trading halts. Blocks order placement immediately, reversible in one click, fully audited.

OMS / Risk Config DONE

Editable global risk limits enforced on every order — starting with quantity and order-value caps, later extended to all 14 controls.

Audit Log DONE

Every admin action — bans, kill switches, role changes, config edits — writes a timestamped, attributable record.

User & Role Management DONE

Role and status edits, extended later with admin-initiated account creation and entity scoping.

PM Shell DONE

Rebuilt as a self-contained, view-only shell — no route into the edit-capable Admin Portal, matching the spec's read-only/approval-routing scope for this role.

Company Account Shell DONE

All 7 previously dead cards now route into the same Admin Portal RMS Admin uses.

Super Admin Shell DONE

Remaining dead cards wired; Company Account Management left for Phase 2 rather than faked.

```
+ new server/src/routes/{kill-switch,oms-config,audit,security-limits}.routes.js
+ new client/src/components/admin/{KillSwitchPanel,OmsConfigPanel,AuditLogPanel,UserRoleManagement}.jsx
```

~ **mod** client/src/pages/AdminPortal.jsx – tabbed shell for every admin module

+ **new** migrations/postgres/02_rms_admin_modules.sql

Company Account as a real entity

Before today, "Company Account" was a role label on a user row — no entity record, and no way for anyone to create an account at all except self-service sign-up.

Companies table DONE

A company is now a first-class record: code, name, status. The organisation boundary the spec describes.

Create / suspend flow DONE

Super Admin creates a company and its first Company Account login in one step, with a one-time temporary password. Suspending a company blocks login for every user under it instantly — and reactivating is just as instant.

Admin-initiated account creation DONE

Previously impossible for any role. Company Account can now create RMS Admin / PM / Trader logins within its own entity only — enforced server-side, not just hidden in the UI.

Entity-scoped user list PARTIAL

A Company Account only sees its own entity's users. Order Book, Positions, Audit Log, and Kill Switch remain platform-wide — a scoped follow-up, not done yet.

```
+ new server/src/{routes,controllers}/company.routes.js / company.controller.js
+ new client/src/components/admin/CompanyAccountManagement.jsx
~ mod admin.controller.js — createUser, entity-scoped getAllUsers
~ mod auth.controller.js — login blocked for suspended companies
+ new migrations/postgres/03_companies.sql
```

All 14 RMS pre-trade risk controls

Spec Section 8, in full. Every order placement now runs this exact gauntlet before it's accepted — plus a separate 120 orders/sec cap and LIMIT-only entry.

#	CONTROL	ENFORCEMENT	STATUS
1	Price Check	Order price banded against live LTP ($\pm$ configurable %)	DONE
2	Quantity Limit Check	Per-order quantity cap	DONE
3	Order Value Check	Per-order price $\times$ qty cap	DONE
4	Trade Price Protection	Combined with Control 1, as spec'd	DONE
5	Market Price Protection	Spec: "not currently enabled" — moot once Market orders removed	N/A BY DESIGN
6	Cumulative Open Order Value	Sum of user's own pending order value	DONE
7	Net Position vs. Margin	New per-user <code>available_margin</code> field	DONE
8	Position Limit Check	Existing + pending + this order, per symbol	DONE
9	Trading Limit Check	Shares the turnover figure below, distinct label	DONE
10	Exposure Limit Check	Both user-level and platform-wide global limit	DONE
11	Turnover Limit Check	Cumulative executed value vs. configured cap	DONE
12	Security-Wise Limit Check	New per-symbol qty/value override table	DONE
13	Automated Execution Check	Caps outstanding pending orders per user	DONE
14	Fair Play Check	Commodity Derivatives-specific; segment doesn't exist here	N/A BY DESIGN

Plus: **OPS cap** — configurable orders/sec rate limiter (default 120) — and **LIMIT-only order entry** with F1 / F2 keyboard shortcuts, per Section 7.1.

HFT-grade grid & WebSocket layer

Section 5's literal ask — "the GUI must feel instant" — implemented, not just aspired to.

Delta protocol DONE

Market data now streams only what changed, with a strict sequence number. A dropped message is detected and self-heals via a resync request — verified with a raw socket client.

Code-split by role DONE

Split one 1.5 MB shared bundle into five per-role chunks — RMS Admin and Company Account now load ~1.8 KB instead of the whole app.

Grid transactions DONE

Watchlist, Order Book, Net Positions, and Ban Script all update via row-level transactions instead of full re-renders.

Grouping PARTIAL

"Group by Symbol" ships as a Community-tier substitute. True OMS/Token grouping needs a paid AG Grid Enterprise licence — a decision for the business, not engineering.

CSV export & columns DONE

Every grid exports to CSV, and column order/width now survives a page reload.

Light theme default PENDING

Spec wants a light, low-repaint default for trading screens; all grids are still dark. Flagged, not started.

Every table created or extended today

Three migration files, applied in order, all on PostgreSQL. Commands used shown below each migration.

```
psql "$DATABASE_URL" -f migrations/postgres/02_rms_admin_modules.sql
psql "$DATABASE_URL" -f migrations/postgres/03_companies.sql
psql "$DATABASE_URL" -f migrations/postgres/04_rms_controls.sql
```

kill_switches

02_rms_admin_modules.sql

Trading halts. Presence of a row = active — same pattern as the existing banned_scripts table. A partial unique index guarantees at most one active GLOBAL row and one per user.

id	uuid, pk	scope	'GLOBAL' 'USER'	target_user_id	fk → users	reason	varchar(255)
activated_by	fk → users	activated_at	timestampz				

oms_config

02_rms_admin_modules.sql → extended in 04

Single-row global configuration table (id always 1). Started with 2 risk fields, grew to 10 as the 14 controls were built out.

max_order_quantity	numeric	max_order_value	numeric	price_band_pct	numeric	max_open_order_value	numeric
max_position_qty	numeric	max_exposure_value	numeric	global_exposure_value	numeric	max_turnover_value	numeric
max_open_orders_count	int	max_orders_per_second	int				

audit_log

02_rms_admin_modules.sql

One row per admin/RMS action platform-wide — ban, kill switch, role change, config update, company create/suspend.

actor_user_id	fk → users	action	varchar(50)	target	varchar(100)	details	varchar(500)
created_at	timestampz, indexed						

companies

03_companies.sql

Makes Company Account a real entity. `users.company_id` gained a foreign key into this table's code.

code	varchar(100), pk	name	varchar(255)	status	'ACTIVE' 'SUSPENDED'	created_by	fk → users
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security_limits

04_rms_controls.sql

Per-symbol quantity/value overrides for Control 12. Presence-based, same pattern as `banned_scripts` — no row means no override for that symbol.

symbol	varchar(50), pk	max_qty	numeric	max_value	numeric	set_by	fk → users
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users, orders – column changes

03 & 04

`users.available_margin` added (Control 7). `orders.type` constraint tightened from MARKET|LIMIT to LIMIT-only; price is now always required.

Full spec coverage, one list

Every requirement from the iwmQT GUI spec, mapped against what's actually built — the way a `requirements.txt` pins a dependency, not a status adjective.

Section 2 – Role Model & Access Hierarchy

five-role-model	[x] done
role-based-routing	[x] done
all-dashboards-wired	[x] done
company-account-entity-model	[x] done
entity-scoping-every-module	[~] partial user mgmt only

Section 3 – Login, Auth & Password Policy

otp-signup-and-reset	[x] done
lockout-5-attempts	[x] done
password-expiry-15-days	[~] partial doesn't reset on change
password-reuse-check	[] pending
forced-password-change-flow	[] pending
idle-session-timeout	[] pending
single-active-session	[] pending

Section 5 – Performance (HFT)

websocket-delta-protocol	[x] done
sequence-numbers-and-resync	[x] done
ag-grid-row-transactions	[x] done
code-splitting-by-role	[x] done
light-theme-default	[] pending
binary-framing	[] pending messagepack/flatbuffers, optional

Section 6 – AG Grid Contract

csv-export-all-grids	[x] done
column-layout-persistence	[x] done
oms-token-wise-grouping	[~] partial needs Enterprise licence

Section 7 & 8 – Order Entry & Pre-Trade Controls

limit-only-order-entry	[x] done
keyboard-shortcuts-f1-f2	[x] done
ops-cap-120-per-sec	[x] done
all-14-rms-controls	[x] done
trade-book-distinct-screen	[] pending
servers-oms-provisioning-screen	[] pending

What's left, in priority order

Two meaningful phases remain, plus a handful of smaller, independently-schedulable gaps.

1 • Auth & session hardening

Password reuse check, forced-change flow, idle-session timeout, single active session. Security-focused, self-contained.

2 • Light theme for trading screens

Spec-mandated default; currently dark everywhere. Small, fast follow-up.

3 • Trade Book & Servers/OMS screens

Two screens the spec names explicitly that don't exist as distinct views yet.

4 • AG Grid Enterprise decision

A licensing call, not an engineering one — needed before true OMS/Token grouping can ship.

Questions

Full source on `feature/dev` — commits `4eec107`, `d2d95cc`, and `634ca3c`. `main` is currently one commit behind and due for a fast-forward merge.